K Disentangling Format-Specific and Abstract Representations in Function Vectors

Multiple-choice (MC) format involves distinct computational steps beyond open-ended generation: evaluating options, comparing candidates, and selecting among labeled alternatives (Tulchinskii et al., 2024; Wiegreffe et al., 2025). When FV heads are extracted from MC prompts, they must therefore capture both (a) the task representation (e.g., antonym, causation) and (b) the MC-specific formatting demands. We test whether partitioning out of the MC information, makes the task representation abstract, i.e., is shared across all formats?

We partition FV heads in Llama 3.1 70B into three subsets:

- **All Heads**: Top-5 heads identified by AIE computed over all input formats (standard FV selection, Eq. 5)
- **Common Heads**: Heads that appear in the top-10 heads for *all three* input formats independently. (3 heads).
- **Unique MC Heads**: Heads that appear in the top-10 heads for MC format only. (6 heads)

If abstract task representations exist in FVs independent of format, we would expect them to reside primarily in the **common heads** that are causally important across all formats. We then computed similarity matrices and RSA scores for each head subset.

We see that within the MC format, common heads cluster by concept (Figure 19), but the representations are nearly orthogonal between open-ended and multiple-choice prompts for the same concept (Figure 20).

It is still possible that these heads could encode both abstract task in open-ended and format features in MC prompts in superposition. However, under the Linear Representation Hypothesis (Park et al., 2024), a shared abstract concept should occupy a consistent linear subspace detectable via cosine similarity. The observed orthogonality across formats implies that any abstract representation is not linearly accessible in a format-invariant way. This suggests that FVs encode tasks at a lower level of abstraction (i.e., 'antonym in MC format') rather than a shared, format-independent concept.

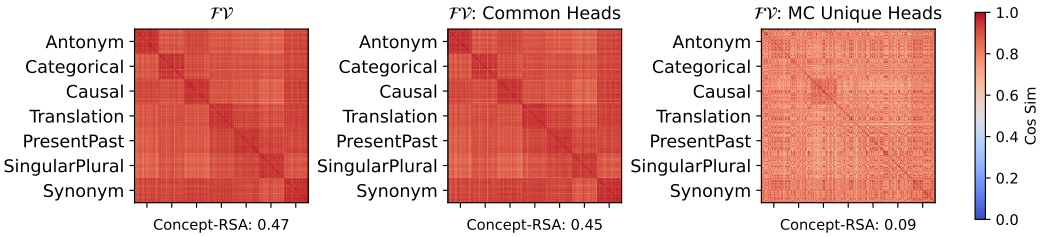

Figure 19: *Similarity matrices for MC prompts only*. Each panel shows the similarity matrix for a different subset of AIE-selected heads computed over MC prompts only (7 concepts $\times$ 50 prompts each). Common heads show stronger concept clustering than unique MC heads (albeit with large similarity between concepts due to the shared MC structure). Full FVs show a very similar representational structure to the common heads. Model: Llama 3.1 70B.

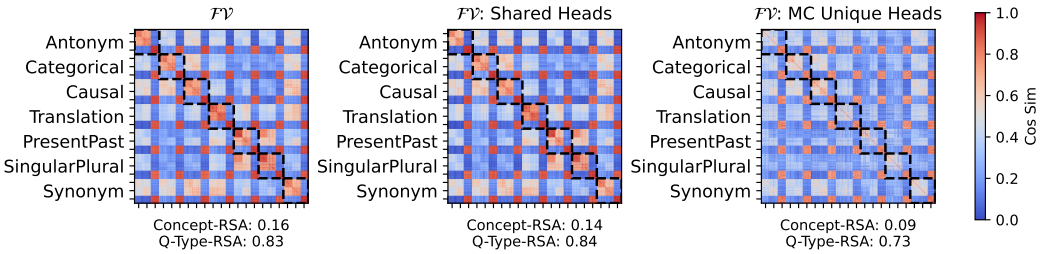

Figure 20: *Similarity matrices for different FV head subsets across all formats*. Same as Figure 19 but computed over all input formats (7 concepts $\times$ 3 formats $\times$ 50 prompts). Within the same concept the representations are nearly orthogonal between open-ended and multiple-choice prompts. Model: Llama 3.1 70B.

L STEERING RESULTS (ACCURACY)

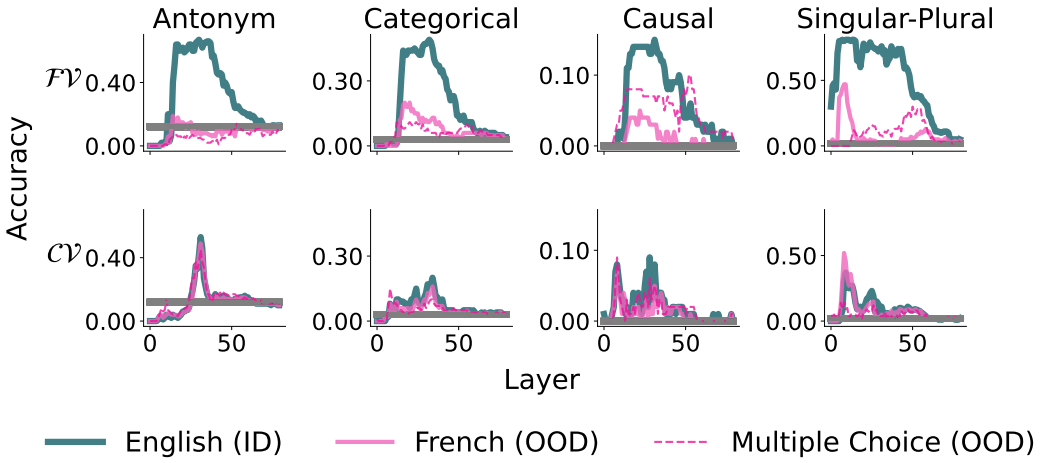

Figure 21: *Steering effect across layers (Accuracy).* We inject $\mathcal{CV}$s and $\mathcal{FV}$s into Llama-3.1-70B and plot the Top-1 accuracy for four representative concepts (columns). The grey horizontal line indicates the accuracy of the unsteered model. See Figure 7 for ΔP results.

M MULTIPLE-CHOICE FORMAT WITH WORDS AS OUTPUT

Example prompt:

```
Instruction: Q: spherical A: ?
unconstitutional
flat
demand
healthy
Response: flat

Instruction: Q: unveil A: ?
optional
mild
sturdy
conceal
Response:
```

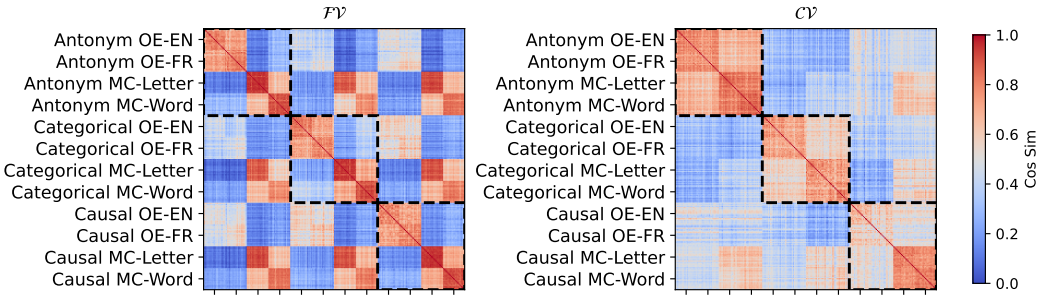

Figure 22: Similarity matrices for $\mathcal{FV}$s and $\mathcal{CV}$s with the inclusion of MC prompts where the model is expected to produce a word instead of a letter. **Takeaway**: Unlike $\mathcal{CV}$s, $\mathcal{FV}$ MC representations still cluster due to the input format, therefore MC cluster effect is not due to the model producing words/letters. Model: Llama 3.1 70B.

N AMBIGUOUSICL IN DIFFERENT LANGUAGES

In the original AmbiguousICL implementation, the concepts are presented in English and the translations are presented in French.

Here, we test the effect of presenting the concepts in a different language to determine if $\mathcal{CV}$s are tied to a specific surface form (e.g., "English Antonym"). Specifically, we present the concepts in Spanish and interleave them with Spanish-to-English translations.

Example antonym prompt:

```
Q: final
A: inicial

Q: inmaduro
A: madura

Q: norte
A: sur

Q: descendiente
A: descendant

Q: probablemente
A: probable

Q: vivo
A:
```

The expected response is the Spanish antonym 'vivo' → 'muerto'. Therefore, the ID vectors are extracted from open-ended Spanish antonym prompts and the OOD vectors are extracted from open-ended English and multiple-choice prompts.

In Figure 23, we see that the steering effect trends are similar to the original implementation (although the absolute performance is lower for both $\mathcal{CV}$s and $\mathcal{FV}$s). In Figure 24, we also see that the KL divergence is lower for the $\mathcal{CV}$s compared to the $\mathcal{FV}$s. Crucially, the $\mathcal{CV}$s (extracted from English) steer the model to produce the **Spanish antonym** (the contextually appropriate response), rather than the English antonym or the English translation. This demonstrates that $\mathcal{CV}$s encode the abstract 'Antonym' concept rather than 'English Antonym'. Mechanistically, this suggests that the $\mathcal{CV}$ amplifies the task probability ("do an antonym") while relying on the prompt's existing context to determine the surface form ("in Spanish"), rather than injecting language-specific content.

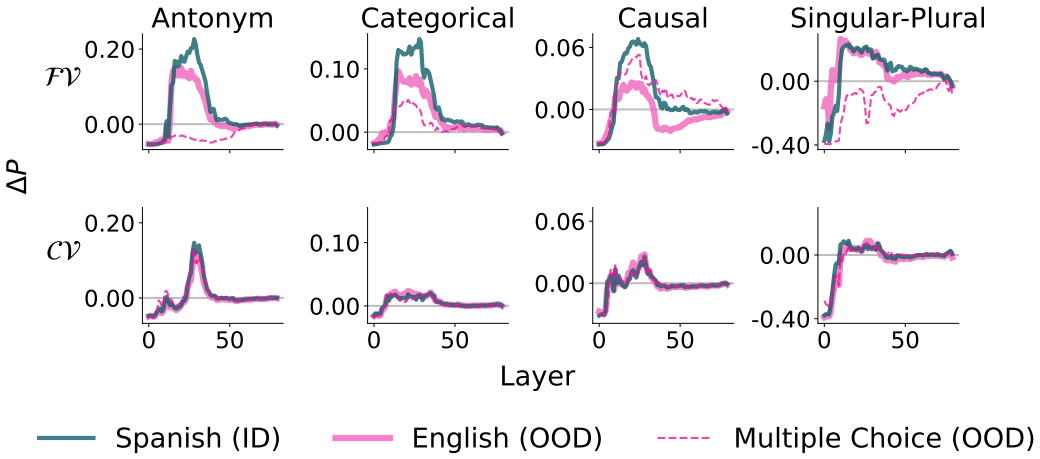

Figure 23: Steering effect across layers and all concepts for different languages.